JIAWEN ZHANG

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EDUCATION

University of Illinois Urbana-Champaign Jun. 2026 – Present
Visiting Researcher, Siebel School of Computing and Data Science *Champaign, IL, USA*

- **Advisor:** Prof. James M. Rehg

Peking University Sep. 2024 – Present
M.S. in Electronic Information, National Biomedical Imaging Center *Beijing, China*

- **Core Courses:** Large Language Models and Applied Practices, Machine Learning, Natural Language Processing, AI in Medicine, Algorithm Analysis and Design, Software Architecture and Design.

Lanzhou University Sep. 2019 – Jun. 2024
B.Eng. in Computer Science and Technology, UAIS Lab *Lanzhou, China*

- **Rank: 1/168 & National Scholarship** (Average Score: 93.36/100)
- **Core Courses:** Introduction to Artificial Intelligence, Machine Learning, Data Structures, Principles of Computer Organization, Computer Networks, Assembly Language.

RESEARCH INTERESTS

Multimodal AI & World Models for human behavior understanding;
Agentic RL and Post-Training of vision–language models;
Computational Imaging and physics-constrained inverse problems.

PUBLICATIONS

- **Jiawen Zhang**, et al. Gaze World Model. Under review at *International Conference on Learning Representations (ICLR)*, 2027.
- **Jiawen Zhang**, et al. Radon–Full-Waveform Inversion for Suppressing Scalp Reverberation and Skull-Induced Aberration in Transcranial Photoacoustic Computed Tomography. *Photoacoustics*, 2025.
- **Jiawen Zhang**, et al. Evaluating Cognitive Age Alignment in Interactive AI Agents. *arXiv preprint arXiv:2605.17894*, 2026.
- **Jiawen Zhang**, et al. Dual-Modality US–PA Aberration Correction for Transcranial Imaging. Under review at *IEEE Transactions on Medical Imaging (TMI)*, 2026.

RESEARCH EXPERIENCE

Rehg Lab, University of Illinois Urbana-Champaign Advisor: **Prof. James M. Rehg** Jun. 2026 – Present
Research Intern Mentors: **Xu Cao**, **Houze Yang** *Champaign, IL, USA*

Gaze World Model (GWM) — Project Lead *Jul. 2026 – Sep. 2026*

- Formulated person-specific gaze modeling in third-person video as a **world prediction problem**, unifying gaze-point estimation, in-clip gaze-shift detection, and forecasting of future gaze targets and shift events.
- Developed **GWM**: a frozen **V-JEPA 2.1** encoder, RoPE-based scene–head cross-attention, a 3D-RoPE temporal trunk, and a latent world predictor trained with a **motion-weighted latent prediction loss** (no pixel reconstruction).
- Built a **human-in-the-loop annotation engine** (rule proposals, multimodal-model verification, human review) that converts frame-level labels in VideoAttentionTarget and ChildPlay into **6,204** event-level gaze shifts and 15,086 training windows.
- Outperformed Gaze-LLE, PaGE, SAGE, Qwen3.8 and Gemini 3.8: shift-detection AP **78.38** → **89.72**, future-shift AP 66.23 → **76.18**, future gaze L2 0.123 → **0.104**; the 105M ViT-B variant beats the 1.7B PaGE ViT-H+ on every shift metric.

PediaMed AI Mar. 2026 – Jun. 2026
Research Intern

ChildAgentEval: Cognitive Age Alignment of AI Agents — First Author *Mar. 2026 – Jun. 2026*

- Built **ChildAgentEval**, a WISC-inspired interactive web benchmark that evaluates the **cognitive age alignment** of multimodal LLM agents across psychometrically grounded cognitive domains.
- Proposed a **two-stage skill-guided distillation** strategy that turns empirical child developmental markers into executable constraints on agent language, memory capacity, and task-solving behavior.
- Evaluated proprietary and open-weight models: cognitive filters yield monotonic, age-ordered score and linguistic progressions, while exposing persistent limitations in calibrating working memory and visuospatial reasoning.

National Biomedical Imaging Center, Peking University Oct. 2024 – Mar. 2026
Graduate Researcher *Beijing, China*

Radon–FWI for Transcranial Photoacoustic Tomography — First Author *Oct. 2024 – Jul. 2025*

- Formulated transcranial PACT as a **PDE-constrained nonlinear inverse problem**, explicitly modeling skull heterogeneity, multipathing, elastic scattering conversion, and scalp-dominated reverberations.
- Proposed a dual-domain **RT–FWI** framework: a high-resolution linear Radon transform separates scalp multiples by slowness (no prior SOS needed), and the resulting masks act as a **scalp-energy regularizer** in elastic FWI, reducing non-uniqueness and cycle-skipping.
- Implemented adaptive Radon-domain masking (block-wise Otsu + morphological cleaning) and 8th-order staggered-grid FDTD with CPML, optimized via adjoint gradients and FISTA with restart.
- Improved axial resolution from **1.14 mm to 0.28 mm** and SSIM from **0.42 to 0.88**.

UMI-PAI: Dual-Modality US–PA Aberration Correction — First Author *Aug. 2025 – Mar. 2026*

- Proposed **UMI-PAI**, a measurement-based framework that estimates skull-induced **phase distortions** from the ultrasound reflection matrix and applies the correction operator to photoacoustic reconstruction at minimal computational cost.
- Developed a spatially adaptive phase-screen model with **iterative reflection-matrix optimization** (adaptive windowing, locally weighted updates), integrated into Fourier-domain PA reconstruction via phase-conjugation compensation.
- Validated on acrylic phantoms and murine skulls: up to **44%** PSNR gain (39.30 dB) over baselines, without CT/MR priors or costly elastic-wave inversion.

INDUSTRY EXPERIENCE

Tencent Holdings Limited Dec. 2025 – Mar. 2026

Research Intern, Multimodal Large Models Leader: **Qi Xiong**

- Distilled 127K structured CoT trajectories from a 397B teacher into a **4B multimodal model** via **multi-task SFT**, fusing text, DOM trees, and visual inputs for generative web security assessment.
- Built an **agentic RL** post-training pipeline with GRPO, **dynamic prompt sampling** (prioritizing medium-difficulty tasks by success-rate variance), and a composite reward on reasoning fidelity, chain density, and length to curb reward hacking.
- Cut tool-call rounds from 5.43 to **2.90** and end-to-end latency by **70.8%**, while raising decision accuracy to **88.50%**, surpassing the teacher model.
- Designed fallback reasoning for crawler failures and visual-input timeouts, reducing false positives by **65.9%** across 221K live traffic samples.

HONORS & AWARDS

2025 **Peking University Social Scholarship**, Peking University
2023 **Finalist (F Award)**, Mathematical Contest in Modeling (MCM)
2022 **National Encouragement Scholarship**, Ministry of Education of China
2021 **National Scholarship**, Ministry of Education of China
2021 **Outstanding Student Model**, Lanzhou University

LEADERSHIP

Organization Commissary, School of Software and Microelectronics, Peking University 2024 – Present
Deputy Class Advisor, Computer Science Cohort 2022, Lanzhou University 2022 – 2023
President, Student Association Management Center, Youth League Committee, Lanzhou University 2021 – 2022
Department Officer, Student Association Management Center, Lanzhou University 2020 – 2021

SKILLS

Languages: English (Fluent), Chinese (Native)
Programming: Python, C/C++, CUDA, Verilog, MATLAB
Frameworks: PyTorch